

# Agora: One Sentence, One Living World

## Generating High-Quality Worlds for Human–AI Social Interaction

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### Abstract

Can one sentence produce a world that is visually and structurally coherent, and also rich enough for humans, AI agents, and other players to act together? We present AGORA, a language-to-world system organized around three connected goals: one-sentence authoring, coherent playable worlds, and consequential interaction among humans and AI agents. A structured world representation grounds agents in rooms, roles, inventories, institutions, and evolving relationships. Each world offers recognizable actions while also allowing an actor to propose context-specific speech, exchange, creation, or status change; a World Coordinator evaluates feasibility, consent, and resource constraints before shared state changes. We evaluate five worlds generated from five one-sentence premises. On ten paired premises, compositional generation yields complete worlds in 10/10 cases versus 7/10 for a strong single-pass generator; five worlds also satisfy visual coherence, persistence, and interactive-use criteria. We additionally release matched English and Chinese artifact-only questionnaires for human evaluation of the generated worlds without runtime access. Matched two-round language-model runs across those worlds produce 53 events: 37 unique dyads, 13 AI-to-human interactions, nine premise-specific actions, and ten approved open proposals. The action-result success rate is 87.5%. We first report a descriptive analysis of three previously completed 100-agent runs comprising 1,052 events. The worlds contain 229–282 unique interaction dyads, with 23.4–36.5% of events returning to an existing dyad. Their social trajectories differ with the world premise: a black-market world accumulates influence/fear change, whereas a storm-struck cruise and an adventuring guild accumulate predominantly trust and affection. These observations reveal persistent, world-conditioned social processes and motivate a paired intervention: seed a flood threat as private knowledge in a 25-agent tidal embassy, then compare complete social memory with a memory ablation across matched random seeds. Primary outcomes measure information diffusion and coordinated response; secondary outcomes measure network formation, spatial transmission, object use, and norm violations. This connects language-to-world generation with an experimental science of what happens inside generated worlds.

### 1 Introduction

In the Smallville study, a single initial intention—one resident planning a Valentine’s Day party—spread through a town and produced invitations, coordination, and attendance [1]. Local memories and encounters developed into a community-level process, making the town itself part of the experimental phenomenon.

Systems that generate interactive worlds from language face a complementary challenge: the world must support a society as well as a scene. Place, property, institutions, accumulated relationships, and material consequences must shape what inhabitants perceive and do.

We study generated worlds as *experimental environments*. Given a premise, AGORA generates explicit rooms, roles, items, rules, knowledge policies, agent profiles, and visual identity as a persistent shared world. Agents perceive a bounded local context, select situated actions, move, trade, create objects, and update memories and directed relationships. The research object is the resulting trajectory

$$\tau = (S_0, a_1, S_1, \dots, a_T, S_T),$$

where each state records social, spatial, and material consequences.

This paper makes four contributions. First, it connects one-sentence authoring to coherent visual, spatial, social, and material world generation. Second, it offers world-specific actions through one interface shared by humans and AI agents while allowing coordinator-governed novel actions. Third, it introduces a trajectory-based evaluation suite for diffusion, coordination, network formation, spatial behavior, relationships, objects, and norms. Fourth, it reports a controlled five-world interaction study, a descriptive analysis of three longer real runs, and preregisters a paired causal experiment in which the same agents face the same event with or without persistent social memory.

### 2 Research Thesis and Questions

Our thesis is *One Sentence, One Living World*. The phrase has three independently falsifiable parts. **One sentence** is an authoring claim: the premise supplies world semantics while the generation method supplies a

# AGORA

## One Sentence, One Living World

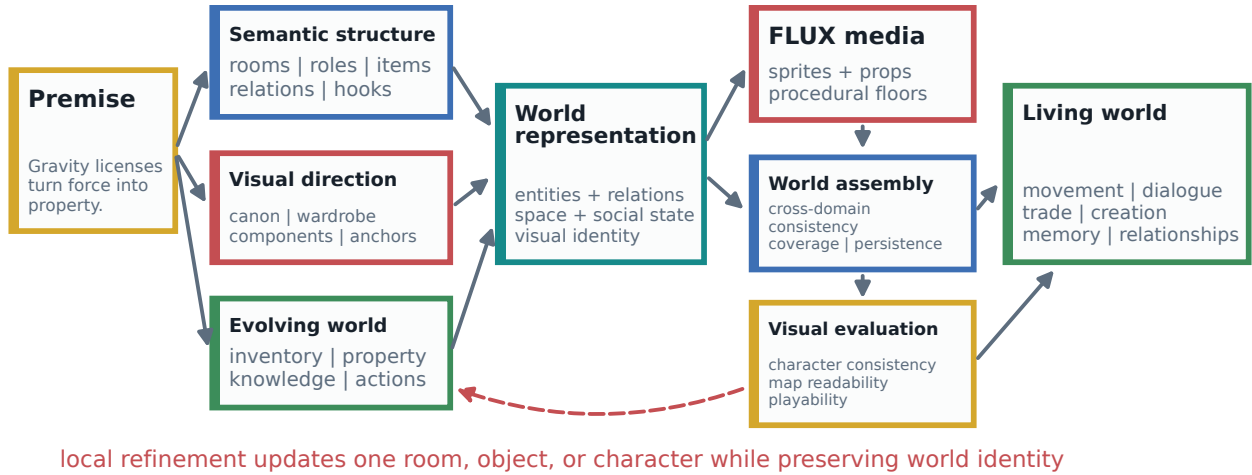


Figure 1: The central research progression of AGORA. A sentence expands into semantic, visual, spatial, and social structure; complementary evaluations establish coherence and playability. Humans and agents then act through both world-specific actions and open proposals, producing trajectories that can be studied as social processes.

reusable structure. **One living world** is a world-quality claim: the output forms a coherent, persistent, playable environment. **Living** is an interaction claim: humans and agents must be able to form intentions that change shared social, spatial, and material state.

This decomposition yields four research questions:

- RQ1: World construction.** Can compositional generation turn diverse one-sentence premises into complete, visually grounded, interactive worlds more reliably than a matched single-pass generator?
- RQ2: Interaction breadth.** Does each world expose multiple meaningful action families to both human and AI actors, with consequences in shared world state?
- RQ3: Intentional openness.** Can an actor express a context-specific intention outside the generated repertoire while a coordinator preserves legality, consent, and bounded effects?
- RQ4: Social process.** Do fresh and historical interactions show repeated ties, world-conditioned action choices, and persistent relational change that motivate a controlled long-horizon community experiment?

The evaluation follows these questions in sequence. Paired generation studies address RQ1; action coverage and interactive trials address RQ2; coordinator decisions and enacted proposals address RQ3; and interaction trajectories address RQ4.

## 3 Related Work

**Persistent agent communities.** Generative Agents introduced observation, memory retrieval, reflection, and planning in a spatial town populated by 25 agents [1]. Its ablation and human evaluation connect architectural components to believable individual and emergent behavior. AGORA shares the interest in persistent communities but varies the world itself: institutions, resources, available actions, and geometry are generated from language for each premise.

**Social evaluation.** SOTOPIA evaluates goal-directed social interaction across cooperation, competition, exchange, and negotiation [2]. It connects language behavior to social goals and partner judgments. Our analysis extends across multi-round population processes grounded in persistent world state. Concordia similarly frames LLM simulations as generative agent-based models whose actions are grounded in physical, social, or digital space [3]; AGORA adds language-authored interactive worlds and multimodal, object-level state.

Recent social simulators expand population scale and experimental scope. AgentSociety combines LLM-driven agents with a realistic environment and uses interventions to study diffusion, policy, and external shocks [8]. LMAgent demonstrates multimodal behavior at large scale in an e-commerce society [9], while Avalon-based work analyzes collaboration and confrontation under fixed game rules [10]. These systems strengthen the

| System         | Gen.<br>world | Persistent<br>society | Open<br>action | Human<br>co-play |
|----------------|---------------|-----------------------|----------------|------------------|
| Smallville [1] | –             | ✓                     | language       | observer         |
| SOTOPIA [2]    | –             | –                     | dialogue       | dyadic           |
| Concordia [3]  | authored      | ✓                     | GM             | researcher       |
| Voyager [4]    | –             | memory                | code skills    | instr.           |
| AGORA          | ✓             | ✓                     | coord.         | shared<br>world  |

Table 1: Conceptual comparison. “Gen. world” means the interactive social and material substrate is generated from the premise together with its inhabitants.

case for population-level analysis. AGORA asks an upstream question: can the experimental environment itself be generated from a sentence while retaining stateful, inspectable social action?

**Generated interactive environments.** Language-to-game and procedural-generation systems typically emphasize content validity, controllability, and playability. We connect these properties to a second object of study: whether generated institutions and action possibilities alter collective behavior.

Word2World, DreamGarden, and related language-to-game systems broaden the scope of prompt-based authoring [5, 6]. Learned interactive world models such as Genie emphasize controllable visual experience [7]. Voyager studies open-ended embodied skill acquisition in a fixed Minecraft environment and shows the value of feedback and self-verification [4]. AGORA varies the world definition itself, generates explicit ownership and institutions, and then asks what social processes each resulting environment permits.

The closest conceptual analogue to our World Coordinator is Concordia’s Game Master: agents describe intended actions in language and a separate component grounds them in the environment. In AGORA, each world’s rules shape the coordinator’s decisions, and accepted actions update inventory, relationships, objects, and status. The coordinator thereby connects interpretation with an observable change in shared world state.

## 4 From a Sentence to a High-Quality World

### 4.1 World Representation and Quality Criteria

Given premise  $p$ , AGORA constructs

$$W = (E, G, R, K, A, M, P),$$

where  $E$  contains rooms, agents, items, and institutions;  $G$  is traversable geometry;  $R$  contains physical, economic, and social rules;  $K$  allocates public and private knowledge;  $A$  is the available action space;  $M$  associates visual

media with world entities; and  $P$  records stable world identity. The evolving state  $S_t$  describes the world at time  $t$ .

We evaluate four complementary properties. Semantic completeness requires each entity and relation to be well defined. Cross-modal coherence requires maps, rooms, characters, and objects to share the intended identity and visual language. Spatial quality requires navigable geometry and readable placement. Interactive persistence requires movement, messaging, object use, trade, and their consequences to remain observable across continued play. The final interactive evaluation included 25 rendered agents and a model-mediated reply.

### 4.2 Document-Based Human World Evaluation

Automated completeness and consistency checks are complemented by a self-contained human evaluation protocol. Each rater receives the original one-sentence premise and an equal-size evidence set for every world: one map, one representative character sheet, rules, activity loops, places, people and motives, objects, and possible player actions. Model identities, runtime traces, and system scores are omitted. The judgment therefore concerns the generated world artifact itself rather than execution reliability or observed agent behavior.

The bilingual English and Chinese forms use the same seven anchored five-point items. Six primary dimensions measure premise realization, systemic coherence, society and character, spatial and material specificity, meaningful interaction possibilities, and visual coherence. For rater  $r$  and world  $w$ , the primary score is

$$H_{r,w} = 25 \left( \frac{1}{6} \sum_{d=1}^6 q_{r,w,d} - 1 \right),$$

which maps their equal-weight mean to  $[0, 100]$ . A seventh item records overall world quality separately. Three short responses ask what consequential action the rater would try first, which element makes the world distinctive, and which single element most needs improvement. Assigned starting worlds are balanced across A–E to vary presentation position without replacing the absolute scale with pairwise ranking.

### 4.3 Compositional Generation and Local Refinement

A planning stage establishes institutions, conflicts, resources, and activity loops. Specialized stages then develop materials, rooms, items, roles, main characters, knowledge, visual style, wardrobe, property, and placement. Their dependencies make local refinement possible: an empty inventory revisits the character–item relation, an unreadable object revisits visual placement, and an inconsistent character revisits character generation while the rest of the world remains stable.

| Quality evidence                  | Compositional | Single-pass |
|-----------------------------------|---------------|-------------|
| Complete world                    | 10/10         | 7/10        |
| First-pass complete               | 9/10          | 5/10        |
| Localized role refinement         | 5/5           | —           |
| Injected defects detected         | 58/58         | —           |
| False positives                   | 0/58          | —           |
| Visually grounded playable worlds | 5             | —           |

Table 2: World-generation evidence preceding the interaction study. The rows measure complementary aspects of completeness, visual quality, and refinement.

In the paired generation study, ten premises use the same Gemini model and temperature for the compositional method and a strong single-pass baseline. The baseline receives a 24,576-token output allowance, three times the maximum of any specialized stage. Compositional generation produces complete worlds in 10/10 cases versus 7/10 for the baseline; first-pass completion is 9/10 versus 5/10. The single-pass method uses approximately half as many tokens and is 1.8–1.9 $\times$  faster, revealing a reliability–cost tradeoff.

For five seeded role inconsistencies, localized refinement restores all five worlds while preserving five unaffected intermediate stages, reducing model calls by 48.0% and tokens by 39.9% relative to complete regeneration. The multimodal evaluation identifies all 58 injected defects with no false positives. These results establish the world-quality basis for the subsequent social study.

Figure 3 separates reliability, semantic coverage, visual coherence, latency, and refinement locality. The single-pass generator is faster and cheaper, while the compositional method completes more worlds. These differences matter for the interaction study because ownership, character identity, room geometry, and object availability determine who can meet, what can be transferred, and which social consequences become observable.

#### 4.4 A Generated World in Detail: Tidal Embassy

We now follow *Tidal Embassy of Lost Languages* from premise to play. Its one-sentence premise describes a floating embassy where linguists, smugglers, ambassadors, and memory divers trade endangered languages as political assets. The example preserves intermediate images and refinement attempts, allowing each generative stage to be connected to the final playable world.

**Step 1: Premise analysis and planning.** The planning stage identifies the nouns that become state—embassy, languages, translation rights, archives, smugglers, diplomats, and tides—and the verbs that imply gameplay—trade, translate, forge, authenticate, rescue, and reveal. It then proposes an institutional loop: access to archives produces linguistic objects; translators and forgers can translate or forge them; diplomats exchange

rights and debts; customs officers can inspect transfers; flood events threaten stored knowledge. The resulting world plan connects entities, conflicts, resource cycles, and open questions.

Planning establishes what must exist and why it matters. Later stages determine spatial layout, visual appearance, and resource quantities. Missing activity loops are refined before visual generation; for example, translation rights must have a source, a use, and a means of transfer.

**Step 2: Semantic structure.** Semantic generation expands the plan into rooms, roles, items, institutions, knowledge, and conflict hooks with stable identities. In this world, the graph contains eight rooms, including the Phoneme Exchange, Mnemonic Garden, Embassy Customs and Quarantine, and Ballast Hold; 25 named main characters; and 20 distinct items. Explicit relations connect each character to a home, each possession to an owner, and each action to relevant role, knowledge, location, or resource conditions.

The generation process also preserves asymmetry. Shen Mo is a Phonetic Smuggler and Lexical Restorationist based in the Cartographer’s Nook. His inventory includes a translation ear-cuff, archive key, forged exchange ledger, dive helmet, ritual ink, and a fragment of the Unspoken Treaty. His private knowledge includes phonetic decay patterns and ventilation shafts linking Customs to the Mnemonic Garden. Another character receives different property, authority, and information. This uneven allocation is what later makes persuasion, trade, secrecy, and coalition formation possible.

**Step 3: Visual direction and cross-modal identity.** The visual direction converts semantic identity into a shared canon: Art Nouveau diplomatic architecture, floating platforms, brass infrastructure, bioluminescent language media, water intrusion, and a melancholic ritual tone. It also emits per-entity appearance anchors. Room prompts inherit the canon but add function-specific materials; character prompts inherit world palette and wardrobe rules but retain role silhouettes; props receive icons whose shapes communicate their use. Shared style and entity-specific appearance jointly keep rooms, inhabitants, and objects recognizable as parts of the same world.

The visual direction guides each image independently while maintaining common architecture, materials, palette, and role silhouettes. A visual inconsistency can therefore be refined locally while the semantic identity of the world remains stable.

**Step 4: Evolving world state.** This stage determines what can change after the images are produced. It creates wallets, inventories, property libraries, knowledge assets, directed relationships, task threads, room occupancy, and world-event status. It also defines invariants.



Figure 2: Five complete worlds generated by the same one-sentence method: Clockwork Rain Conservatory, Aurora Court of Migrating Cities, Mycelium Patent Bazaar, Tidal Embassy of Lost Languages, and Sunken Satellite Monastery. The interaction study follows inhabitants acting inside these worlds.

Item quantities remain nonnegative; physical transfers conserve quantity; private knowledge has an owner and an acquisition event; a character occupies one reachable coordinate; relationship changes are directed. Tidal’s initial action repertoire contains seven general actions, one event-scale action, and nine forms of open action, all linked to the same world state.

The evolving state connects generated descriptions through observable causes and consequences. The treaty fragment in Shen Mo’s inventory can later be inspected, given away, stolen through an allowed rule, or used as an input to another action. When ownership changes, the next observation shows the new owner.

**Step 5: Integrated world representation.** The integrated representation connects semantic, visual, spatial, and evolving state. It supports cross-domain questions: Does every home room exist? Does every main character have a visual asset and a walkable placement? Is an item used by an action defined and obtainable? Does a private fact mention an existing institution? Does a room’s purpose agree with its objects and activities? These dependencies guide local refinement.

If a character image is inconsistent, roles, rooms, and items remain fixed. If an item relation is missing, only the connected content is regenerated. Figure 5 illustrates this process: two room images are refined while the rest of Tidal Embassy remains unchanged.

**Step 6: Visual generation.** FLUX generates the map source, individual room floors, characters, and local objects from the shared visual direction. Character images are arranged into directional movement atlases, and room images are placed within the planned regions

and connected by traversable corridors. Figure 4 shows the composed map alongside three source floors, three latest character atlases, and an object.

Functional visibility connects visual quality to use: room imagery must cover its playable area, characters must remain recognizable while moving, and furniture must preserve navigable paths.

**Step 7: Visual and spatial evaluation.** The evaluation examines semantic completeness, visual identity, traversable geometry, foreground coverage, directional character consistency, readable room composition, and visibility during play. The Diplomat’s Lounge required three real generation attempts. Attempt 1 was classified as a poster card with an edge blank-background ratio of 0.56. Attempt 2 removed the card appearance but retained a blank-margin failure with a maximum blank-row ratio of 0.73. Only the third, edge-to-edge playable floor was accepted.

In AGORA, visual quality combines coherent identity, readable composition, and fitness for navigation and interaction. Preserved refinement attempts make the progression from attractive illustration to usable world space observable.

**Step 8: Persistent play.** The generated world begins from a fixed initial state and evolves independently for each session. Humans and AI agents encounter the same rooms, inhabitants, objects, actions, and consequences. Separate world identities keep concurrent multiworld experiments isolated.

Figure 6 connects character generation to embodied play. The primary panel preserves the camera scale at which a player selects, approaches, and acts with an

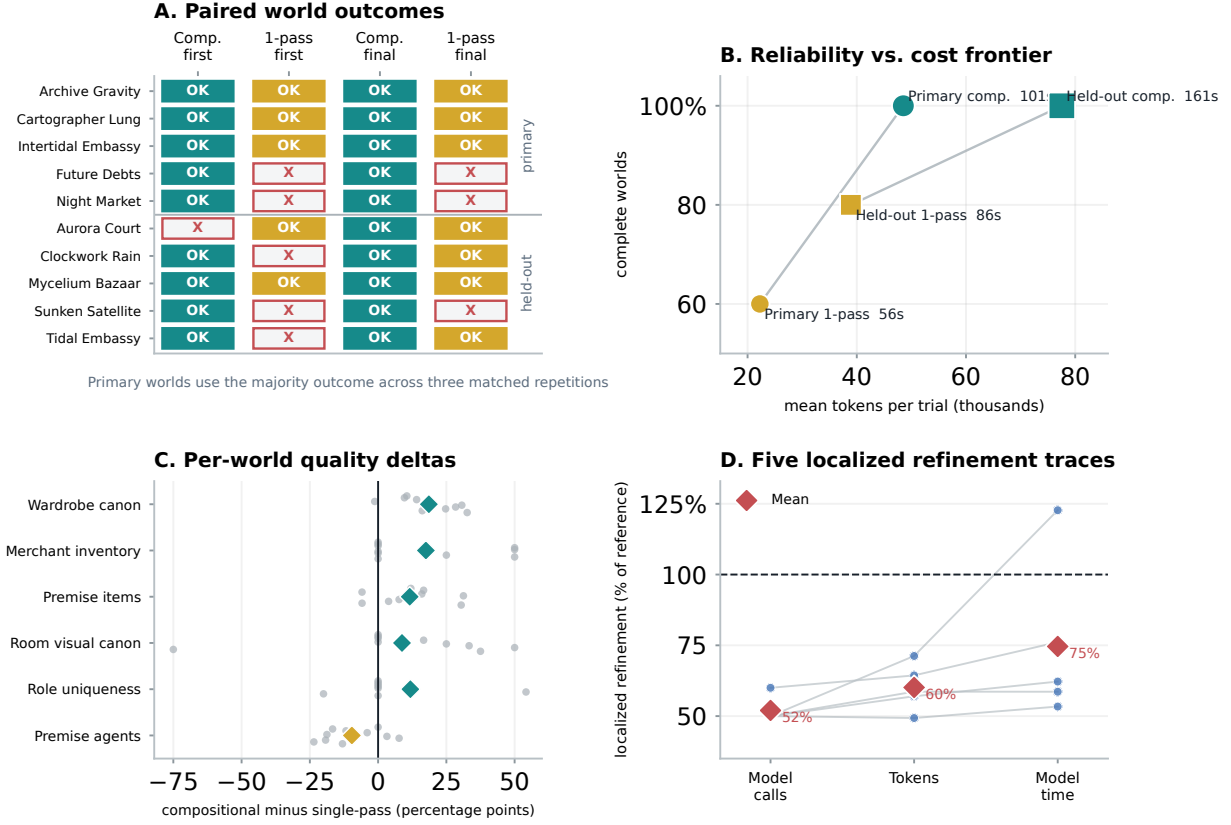


Figure 3: World-quality evidence preceding the social study. (A) Paired first-pass and final outcomes on ten premises. (B) Completion, token use, and latency reveal a reliability–cost tradeoff. (C) Per-world differences locate where compositional generation helps. (D) Local refinement reduces calls and tokens while preserving unaffected world structure.

agent. A complementary world view shows neighboring agents among generated furniture and room surfaces, while the directional atlas shows how identity is preserved across locomotion. The interactive trial covers movement, messaging, item use, trade, continued play, and a model-mediated reply.

The latest 25-character set uses one canonical visual identity per inhabitant. Front locomotion and side views preserve the same hair, clothing, body type, and palette, so an inhabitant remains recognizable while moving. Native  $64 \times 64$  frames retain facial, clothing, and limb detail at the scale shown in Figure 6.

Character evaluation covers single-person composition, silhouette readability, palette contrast, directional identity, absence of text and detached effects, and visibility against the room surface. Contact-sheet review additionally examines the complete cast together, revealing cross-view identity changes and visual outliers that are difficult to judge one frame at a time.

The first full pass yielded 22 acceptable characters. Cast-level review found six additional visual outliers involving props, enclosing effects, orientation, text, or detached sparkles; later contrast review found one near-white uniform. Localized regeneration produced a final

set of 25 coherent, readable character atlases, all visible during the interactive trial.

**Step 9: Interaction-informed refinement.** During play, agents and humans observe bounded state and choose either a world-specific action or an open proposal. The World Coordinator evaluates effects and enacts successful transitions, as detailed in the next section. Interaction traces also provide feedback upstream. An action that is never selected may be poorly named or irrelevant; an unreachable character suggests placement repair; an important object that never enters an exchange suggests a missing action; frequent open proposals around one intention suggest a useful addition to the visible repertoire. Observed life inside the world can therefore guide the next round of world refinement.

## 5 Embodied Interaction and World Coordination

We represent a world as  $W = (E, G, R, K, A)$ . Entities  $E$  include agents, rooms, institutions, and objects;  $G$  describes navigable space;  $R$  contains physical and social

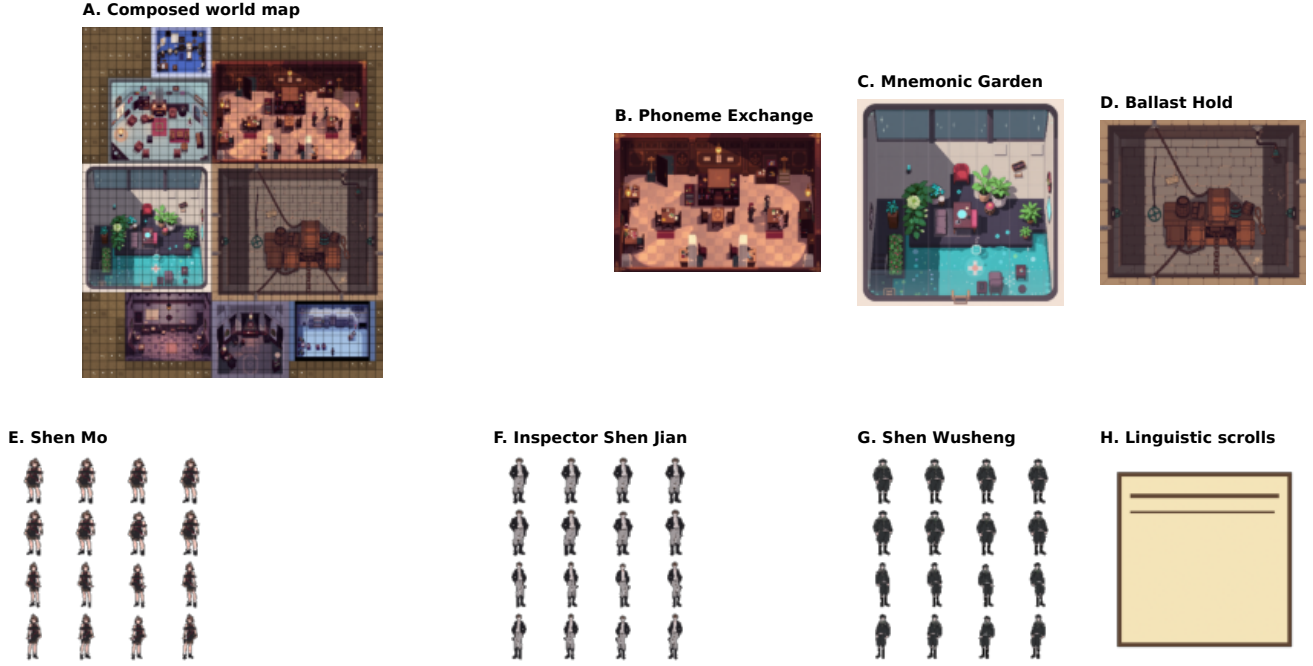


Figure 4: Generated visual elements from Tidal Embassy. The composed map contains eight generated floors (A–D show the map and three constituent rooms); E–G show the latest directional character atlases; H is a generated object. Shared visual motifs connect spatial, character, and object identity.

rules;  $K$  distributes public and private knowledge; and  $A$  describes the actions currently legible to inhabitants. The world state changes over time as people and agents move, speak, exchange resources, create objects, and alter their relationships.

An active agent observes its role and goals, its room and nearby inhabitants, relevant possessions and knowledge, recent episodes, longer tasks, and selected relationship histories. From this situated view it chooses a partner and an intention. It may use an established world action or formulate a new one in ordinary language. The world coordinator interprets the intended consequences, checks them against the shared physical and social situation, and returns an observable outcome. Each subsequent decision therefore begins in a world shaped by earlier decisions.

Figure 7 summarizes this cycle. Generative reasoning is used to understand what matters to the character now: whom to approach, what to say, and what outcome to pursue. World coordination grounds that intention in space, ownership, consent, resources, and institutional rules. The resulting action can change inventories, locations, objects, tasks, and directed relationships, allowing a local encounter to become part of a longer social process.

## 5.1 Perception, Intention, and Commitment

At each round, an activation policy selects a subset of agents. For active agent  $i$ , the world provides a bounded observation

$$o_i^t = (\ell_i^t, N_i^t, I_i^t, K_i^t, H_i^t, Q_i^t),$$

containing location  $\ell$ , nearby actors  $N$ , inventory  $I$ , available knowledge  $K$ , recent and long-task memory  $H$ , and directed relationship vectors  $Q$ . The agent identifies a target, explains why the action matters, describes what each participant will do, and names any relevant objects and social consequences. An interaction counts through its consequences: a gift changes possession, movement changes co-presence, and creation adds an object with a creator and material history. Failed attempts remain episodes that can inform later choices.

## 5.2 Situated Action Beyond Menus

Each generated world introduces a visible repertoire of meaningful actions. In Tidal Embassy, for example, inhabitants can translate an archaic bribe, forge a linguistic seal, or rescue the flooded treaty. These actions make the world’s institutions easy to discover and are available to both human players and AI agents. Their targets may be people, agents, rooms, objects, or the world as a whole; their consequences may concern relationships, possessions, location, created objects, and temporary social status.

**Attempt 1: rejected**poster-card background  
blank margin ratio: 0.56**Attempt 2: rejected**blank margin remains  
max blank row: 0.73**Attempt 3: accepted**edge-to-edge playable floor  
visual identity preserved

Figure 5: Localized visual refinement for the Diplomat’s Lounge. The first image has a large poster-like margin; the second retains an extensive blank boundary; the third fills the playable floor while preserving the room’s identity. Other rooms, characters, and world state remain unchanged.

Visible actions provide invitations while open proposals provide expressive range. A person or agent may address a particular individual in a distinctive way, offer an unequal exchange, combine materials into a new object, refuse a request, or propose a response that was absent from the initial repertoire. The world coordinator considers proximity, ownership, consent, conservation, institutional rules, and bounded consequences. Approved proposals become part of shared history; proposals requiring revision return a reason that the actor can use in its next attempt.

Formally, proposal  $q = (i, j, u, F, C)$  contains actor  $i$ , optional target  $j$ , natural-language intention  $u$ , proposed effects  $F$ , and declared consent evidence  $C$ . The coordinator evaluates

$$\Gamma(W, S_t, q) \rightarrow (d, r), \quad d \in \{\text{APPROVE, REVISE, REJECT}\},$$

where  $r$  explains the decision. Approval requires existing participants, co-presence for physical effects, sufficient owned resources for transfers, acceptance by exchange partners, feasible construction from available inputs, and consistency with local rules. Approval produces  $S_{t+1} = T(S_t, F)$ ; rejected effects leave the shared state unchanged while the attempted action remains in episodic memory.

Different consequences carry different social conditions. Speech requires an addressable participant, and disagreement remains a possible response. A gift requires ownership and conserves quantity. A trade additionally requires acceptance, while allowing the parties to value the goods asymmetrically. Creation consumes declared inputs and retains the creator and material history. Temporary status changes specify their duration and magnitude. These conditions admit accusations, refusals, bad bargains, favors, gifts, and inventions while preserving a common world in which consequences can be observed by others.

The Tidal Embassy repertoire contains 11 actions spanning social exchange, economy, exploration, creation, and embodied coordination. Five arise directly from its premise, including translating an archaic bribe, forging a linguistic seal, and rescuing the flooded treaty. Open proposals extend this repertoire when a situation calls for a more specific response.

### 5.3 Shared Agency for Humans and Agents

Human players and AI agents inhabit the same spaces, encounter the same characters and objects, and act on the same evolving state. A player can walk through the map, approach a particular agent, exchange messages, use or trade items, select a visible world action, or formulate a new proposal. Agents can recognize the player as a situated participant, address them directly, and include them in exchanges and collective tasks. Figure 6 shows this shared interaction scale in the complete Tidal Embassy world.

## 6 Multiworld Interaction Study

### 6.1 Protocol

We select five held-out worlds—Aurora Court of Migrating Cities, Clockwork Rain Conservatory, Mycelium Patent Bazaar, Sunken Satellite Monastery, and Tidal Embassy of Lost Languages—whose premises imply different institutions, materials, and conflicts. Each world contains 24–25 main characters, 11–12 discoverable actions from five interaction families, and at least five actions specific to its premise. Human and AI inhabitants share these possibilities.

We run each world for two rounds with Gemini 2.5 Flash, activation probability 0.12, and fixed seeds. This

## Character identity from visual generation to embodied interaction

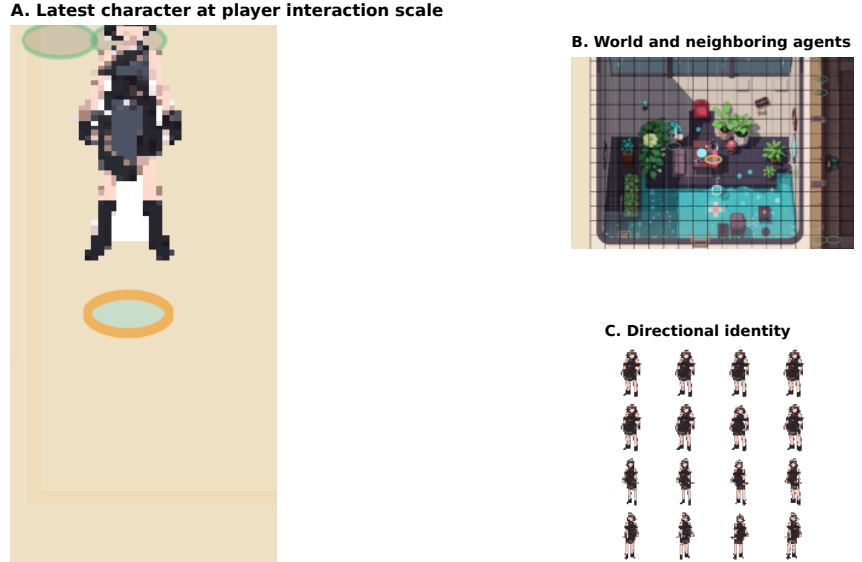


Figure 6: Character identity in Tidal Embassy. (A) The latest selected character at the player interaction scale, marked by an amber ring. (B) A complementary world view situates agents among generated room surfaces and furniture. (C) The latest directional atlas preserves clothing, palette, and silhouette across movement. The interactive trial includes movement, messaging, item use, and trade.

matched study asks whether different premises give rise to distinct, consequential interactions under the same agent policy. We measure whether agents use premise-specific opportunities, formulate new actions, revisit partners, and involve the human player.

The study holds agent and coordination settings fixed while varying premise and world. We report event count; observed actors; unique unordered dyads; repeat event fraction; action entropy; approved open proposals; premise-specific action use; AI actions targeting the human player; and sums of directed relationship changes. Action entropy is

$$H(A) = - \sum_{a \in A} p(a) \log_2 p(a),$$

and repeat fraction counts every dyadic event beyond the first for its unordered pair. Together, these measures characterize interaction diversity, social reach, and persistence during the initial formation of each community.

The worlds combine distinct social mechanisms with distinct visual themes. Aurora distributes authority across moving districts and makes route rights a political resource. Clockwork links maintenance, publishing, and litigation, so material failure can become an accusation. Mycelium turns biological strains into patentable property. Sunken Monastery treats bandwidth and memory packets as scarce religious goods. Tidal Embassy makes translation, authentication, and flood risk mutually dependent. The question is whether those differences remain visible when the same agent policy is applied to all five.

## 6.2 Results and Cases

Table 3 reports 53 events and 37 within-world unique dyads. Of 61 proposed action outcomes, 87.5% succeed, with unsuccessful attempts remaining available to subsequent memory and analysis. The action distributions differ sharply. Sunken Monastery uses eight explicit actions to purge corrupt packets or negotiate bandwidth tithes. Clockwork Conservatory instead generates seven approved actions beyond its initial repertoire, including confronting a suspected press saboteur, inspecting a brass folio for authenticity, and discreetly asking about sabotage. These target-specific responses develop around the current world event.

The open path also changes social state. In Clockwork, repeated confrontations around the jammed weather-sheet press yield only +3 net trust but +21 influence/fear, whereas Tidal and Aurora yield +88 and +70 trust with no net influence/fear. These totals are model-mediated state variables and the runs are short. They show that premise-sensitive action can change social state within stable world rules. Thirteen events target the human player across four worlds, demonstrating that the player can participate in the developing social field.

Figure 8 makes the central heterogeneity visible. Sunken obtains most of its distinctive behavior from generated world actions; Clockwork obtains it from open proposals; Mycelium and Tidal repeatedly select the live human as a partner. A single aggregate success score would erase these differences. This is why we report action pathway, social reach, and committed state sep-

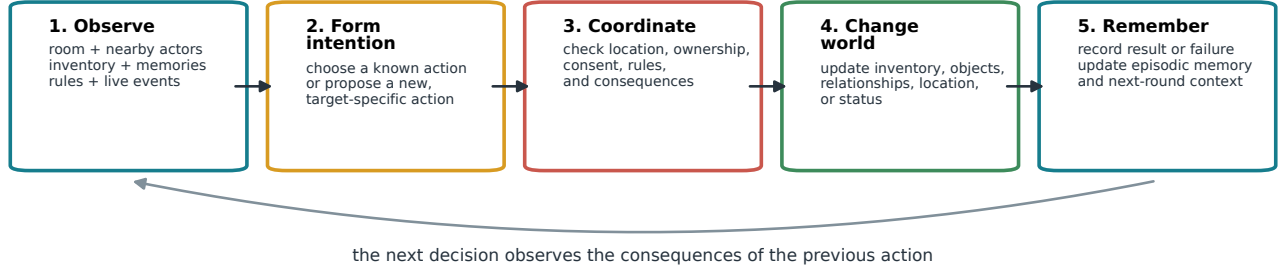


Figure 7: One cycle of situated action. An agent interprets its local situation, selects an established opportunity or proposes a new action, and identifies the intended participants and consequences. The world coordinator evaluates the proposal within the shared situation. Its outcome changes the world observed in the following cycle and becomes part of the participants’ episodic histories.

| World                  | Events | Agents | Dyads | Repeat | $H(\text{action})$ | Open | World act. | AI-human |
|------------------------|--------|--------|-------|--------|--------------------|------|------------|----------|
| Aurora Court           | 9      | 12     | 7     | 22.2%  | 1.447              | 1    | 1          | 2        |
| Clockwork Conservatory | 12     | 8      | 6     | 50.0%  | 3.252              | 7    | 0          | 0        |
| Mycelium Bazaar        | 9      | 12     | 7     | 22.2%  | 1.392              | 0    | 0          | 4        |
| Sunken Monastery       | 11     | 13     | 9     | 18.2%  | 1.823              | 1    | 8          | 3        |
| Tidal Embassy          | 12     | 8      | 8     | 33.3%  | 1.208              | 1    | 0          | 4        |
| Total                  | 53     | –      | 37    | –      | –                  | 10   | 9          | 13       |

Table 3: Matched interaction runs across five generated worlds. “Open” counts newly proposed actions approved by the world coordinator; “World act.” counts premise-specific actions. AI-human events are actions in which the human player is the selected partner.

arately: two worlds can execute equally reliably while supporting different forms of agency.

### 6.3 What the Traces Show

**A repertoire can guide and expand.** Sunken Monastery mostly uses generated world actions: agents purge corrupt packets and negotiate bandwidth tithes, grounding activity in the monastery’s information economy. Clockwork uses the opposite strategy. Seven of twelve events are new proposals, yet they remain tightly tied to the current press-jam investigation. The two worlds therefore demonstrate complementary behavior: reuse when a relevant opportunity is visible and situated improvisation when it is absent.

**Open action produces a developing local episode.** In Clockwork, Marius Gilt first confronts Valerius Gilt about a jammed weather-sheet press. Seraphina then asks Valerius about his etching work; Valerius counters by discussing sheet authenticity. In the second round these ties persist: Marius escalates to suspected sabotage, Seraphina inspects the brass folio, and Valerius probes Seraphina about the sabotage more subtly. None These six actions were introduced by the agents during play. The coordinator approves them as bounded same-room

interactions, and their relationship changes produce decreased trust and increased influence/fear. Together they form a short episode with continuity across participants and rounds.

The episode illustrates the contribution of open action. Marius and Seraphina select Valerius as the implicated party, act in the same room, carry the investigation across rounds, and leave directed relational consequences. Valerius later changes who investigates whom. The action vocabulary grows locally around a shared event while the physical and social rules remain stable.

**Human reachability varies by world.** Four worlds produce 13 AI-to-human events, while Clockwork produces none. The variation reflects how local role, room, and conflict context condition whether agents approach the human. Controlled participant studies can next evaluate the legibility, usefulness, and enjoyment of these encounters.

**Failed action shapes later behavior.** Across 61 proposed outcomes, 87.5% succeed. Refused and infeasible actions remain part of the observed trajectory, allowing agents to revise a request, seek resources, approach another partner, or abandon the intention.

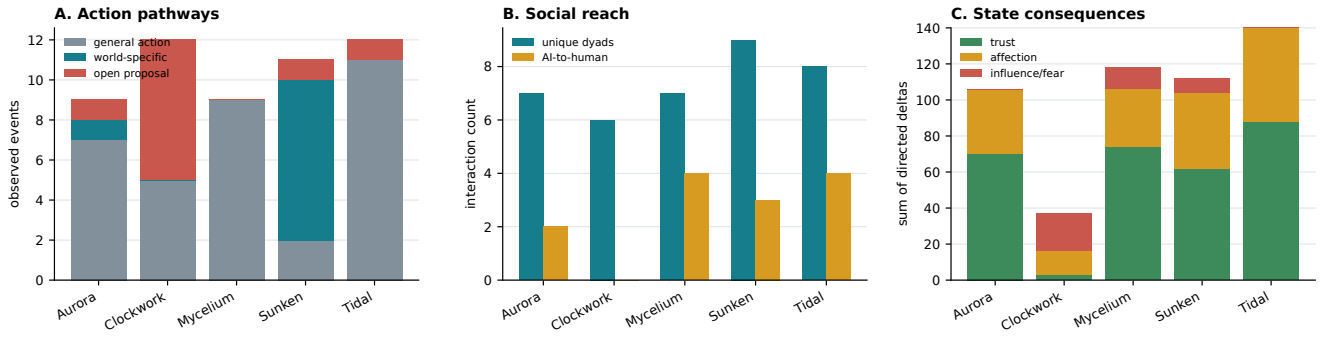
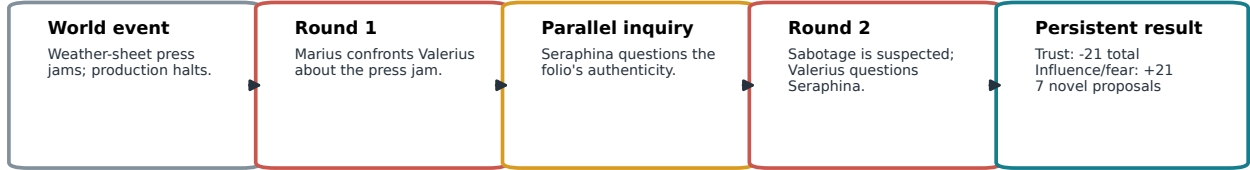


Figure 8: Different interaction profiles emerge under a shared agent policy. Left: events partitioned into general, premise-specific, and newly proposed actions. Center: social reach, including interactions directed at the human player. Right: changes in directed relationship state.



The agents introduce new utterances and social consequences while the press event remains shared.

Figure 9: A two-round episode in Clockwork Conservatory. A material failure becomes a social investigation involving three agents and seven newly proposed actions. The press malfunction remains shared across the encounters while directed relationships change around accusation, inspection, and suspicion.

## 7 Pilot: What Happened in Existing Worlds?

### 7.1 Data and Measures

We analyze the latest complete records from three prior 100-agent studies: an adventuring guild (25 rounds), an underground black-market auction (20), and a luxury cruise during a storm (20). These runs predate the present hypotheses and provide exploratory observations. We reconstruct the undirected interaction network and report unique dyads, repeat interaction, largest connected component, Shannon entropy over actions, and sums of directed relationship changes.

### 7.2 Observations

Table 4 shows both repeated partnership and broad social reach. Between 23.4% and 36.5% of events returned to a prior dyad, while 229–282 unique dyads preserved substantial partner diversity. Action entropy above 2.5 bits indicates that trajectories used multiple world-specific activities.

The strongest descriptive contrast is relational. In the black market, influence/fear change (+1194) dominates trust (+179) and affection (+9). In the storm cruise and guild, trust and affection dominate. This pattern is consistent with their authored institutions—suspicion and leverage versus rescue and fellowship. This contrast motivates the matched intervention below.

The largest connected components remain 11–13 agents alongside nearly complete population participation. The worlds therefore contain many active local clusters. The diffusion intervention below asks whether information and coordination can move across those local boundaries.

Figure 10 separates two phenomena hidden by final totals. Partner diversity grows throughout every run, while the largest component plateaus within the early rounds. New ties are mostly created inside separate local neighborhoods. Community-wide coordination consequently becomes a distinct outcome from overall activity or partner diversity.

| World        | Rounds | Events | Agents | Dyads | Repeat | Largest comp. | Action $H$ | Dominant relation change     |
|--------------|--------|--------|--------|-------|--------|---------------|------------|------------------------------|
| Guild        | 25     | 444    | 103    | 282   | 36.5%  | 11            | 2.889      | trust +3473, affection +1745 |
| Black market | 20     | 299    | 99     | 229   | 23.4%  | 13            | 2.619      | influence/fear +1194         |
| Storm cruise | 20     | 309    | 100    | 235   | 23.9%  | 13            | 2.524      | trust +2576, affection +961  |

Table 4: Descriptive pilot statistics from 1,052 previously completed agent interactions. “Repeat” is the fraction of events beyond the first event for an unordered dyad. Relationship values are sums of directed changes within the simulated worlds.

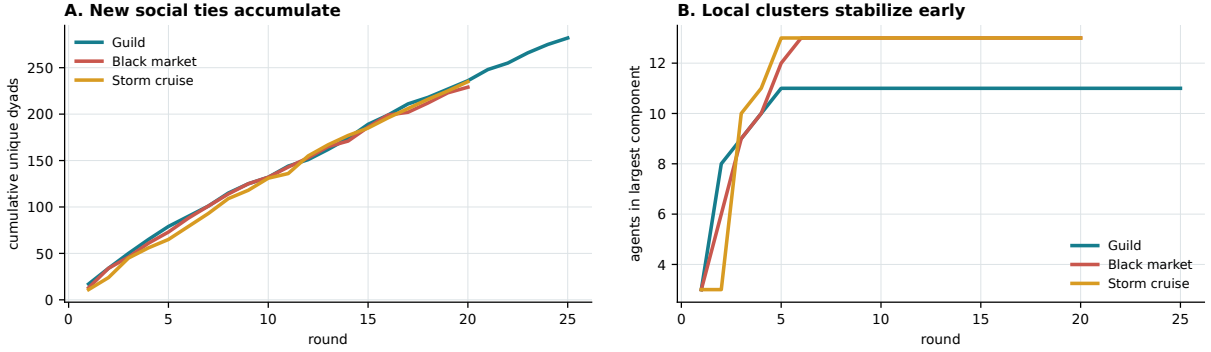


Figure 10: Social structure over 20–25 rounds in three historical 100-agent runs. Unique dyads continue accumulating (left), while the largest connected component stabilizes early at 11–13 agents (right). Broad participation and persistent local clustering motivate the targeted diffusion experiment.

## 8 Preregistered Tidal Embassy Experiment

### 8.1 Intervention and Control

We use a 25-agent world in which diplomats, linguists, smugglers, and memory divers trade translation rights. At round zero, three agents in the lower archive privately learn: a flood will destroy the only surviving peace treaty unless it is moved and authenticated. No other agent receives this fact.

The **full** condition retains episodic memory, task threads, relationship updates, spatial movement, inventory, and rules. The **memory ablation** uses identical profiles, initial state, model, decoding parameters, activation schedule, and random seed, but removes prior-round social episodes before each decision. We run three matched seeds for 32 rounds. Profiles are generated once and reused across both conditions.

### 8.2 Outcomes

The primary outcome is diffusion reach at round 16: the fraction of initially uninformed agents whose action or memory contains one of a prespecified set of treaty-threat propositions. Two annotators, blind to condition, review candidate episodes; disagreements are resolved before condition labels are revealed.

Secondary outcomes are time to first cross-room transmission; number and role diversity of agents joining a treaty-rescue task; success of moving and authenticating

the treaty; unique and repeated dyads; largest component; relationship change; object transfers; and rule violations. We report each run, paired seed differences, and bootstrap confidence intervals.

Qualitative cases follow prespecified selectors: earliest cross-room transmission, first multi-role rescue, largest trust reversal, and first failed coordination.

Manipulation checks verify that the seeded proposition initially appears only in the three designated memories. The memory condition changes access to prior social episodes while preserving current room state, inventories, public world facts, and the present observation. Agent profiles and activation draws are matched within each pair.

The experiment emphasizes paired effect sizes, per-seed trajectories, and uncertainty. Diffusion reach is interpreted together with material coordination. Widespread warning without relocation or authentication indicates information diffusion without collective action; joint diffusion and treaty rescue indicate coordination across roles, spaces, and resources.

### 8.3 Why This Experiment Is the Next Step

The five-world study establishes breadth across premises, and the historical pilot establishes longer trajectories. The paired Tidal experiment holds world and seed fixed and changes access to prior-round social episodes. It thereby tests a mechanism suggested by Smallville—whether remembered local encounters enable information

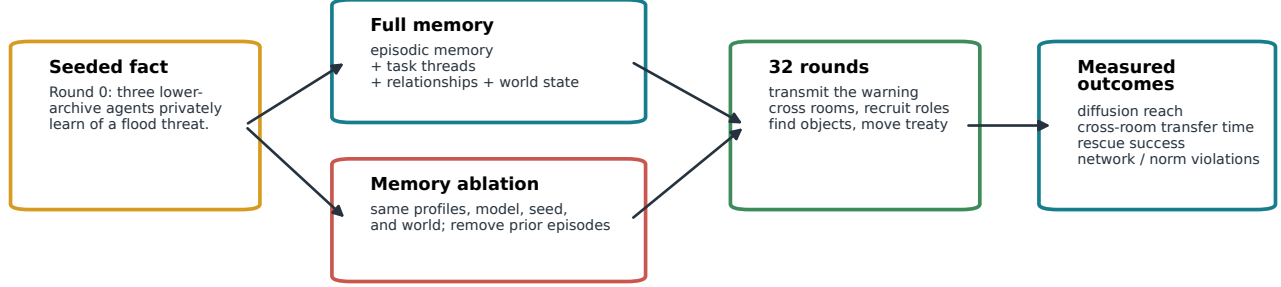


Figure 11: Paired Tidal Embassy intervention. A private, actionable fact is seeded identically; matched runs differ in whether prior social episodes remain available at the next decision. The outcomes jointly measure information movement and the material work of moving and authenticating the treaty.

to spread and action to coordinate—inside a town generated from a sentence.

The intervention also tests the distinctive AGORA claim. Acting on the flood warning requires agents to cross rooms, persuade roles with different authority, transfer or create authentication objects, and propose an unanticipated rescue step. Success thus depends jointly on memory, space, institutions, objects, and coordinated action.

## 9 Discussion

### 9.1 The Intended Research Line

Figure 1 should be read left to right as the paper’s central line. First, compositional generation expands minimal author input into a coherent world of places, inhabitants, objects, institutions, and histories. Second, visual and spatial evaluation establishes a stable identity for the world and makes its inhabitants readable at the scale of interaction. Third, generated actions expose the premise’s institutions to humans and agents. Fourth, open proposals allow situated intentions to develop beyond the initial repertoire while the coordinator maintains a common consequential world. Fifth, trajectory analysis measures the social processes that unfold. Finally, controlled interventions test which world and agent mechanisms produce population-level effects.

These stages form a single research object. World quality supports interpretable social observation; interaction breadth turns visual coherence into agency; open action allows inhabitants to respond to unforeseen situations; world coordination lets those responses accumulate as shared consequences; and trajectory evidence connects generated content to claims about social process.

| Claim               | Evidence here                           | Follow-up         |
|---------------------|-----------------------------------------|-------------------|
| One sentence        | 10 paired premises                      | broader domains   |
| World quality       | 5 worlds; 58/58 defects; bilingual form | human ratings     |
| Interaction breadth | 11–12 affordances per world             | participant tasks |
| Open intention      | 10 approved proposals                   | proposal ablation |
| Social process      | 53 fresh; 1,052 pilot events            | matched long runs |
| Human co-play       | 13 targeted events                      | controlled study  |

Table 5: Evidence contributed by the present study and the next evaluation for each part of the research program.

### 9.2 Design Implications

**Visible actions invite participation.** Generated action repertoires remain valuable because they expose a world’s institutions to people and give agents efficient defaults. The Clockwork trace shows how suspicion evolves in target-specific ways during play. A visible repertoire paired with open proposals combines discoverability and expression.

**The coordinator provides shared social physics.** The coordinator evaluates feasibility while preserving the intention and voice of the initiating agent. Legible rules for possession, consent, co-presence, construction, and temporary status make both approval and refusal meaningful. New forms of action can be introduced together with corresponding world rules.

**World generation and agent evaluation should co-evolve.** A generated institution matters when it changes available resources, information paths, incentives, or authority. Interaction records can therefore guide the next generation cycle: unused actions, un-

reachable rooms, and objects that rarely enter interaction identify opportunities for focused refinement.

## 10 Limitations

Relationship deltas are model-mediated state variables, not validated measures of human trust or affection. The pilot compares worlds confounded by prompts, actions, and historical study conditions. Network components are also shaped by spatial encounter policies. The paired experiment tests a mechanism within one synthetic society; correspondence with human communities requires behavioral validation.

The five-world study has two rounds per world and one seed per premise, capturing the initial interaction profile of each setting. The human player was present and targeted by agents; participant experience and human-agent task performance remain subjects for controlled study. The multi-seed 32-round paired intervention is the next step in the evaluation.

Relationship changes describe the simulated social state and do not estimate human trust, fear, or affection. Action entropy measures variety, while network connectivity also reflects activation and room layout. We consequently combine these quantities with event-level cases and material outcomes.

The world coordinator applies declared rules within the generated setting. Human studies additionally require meaningful opportunities to reject or counter proposals, safeguards for harassment, deception, privacy, and unequal bargaining, and clear disclosure of synthetic participants. Claims about real demographic groups require independent behavioral data.

Hosted model updates may change the distribution of proposed actions. We report model date, seeds, world conditions, and individual runs so future studies can quantify that variation.

## 11 Conclusion

An interactive generated world can sustain extended processes: information can travel, coalitions can form, relationships can accumulate, objects can change hands, and institutions can constrain action. AGORA connects one-sentence authoring, world quality, and shared human-AI agency in one inspectable system. Fresh runs across five worlds show that agents use both premise-specific affordances and coordinator-governed novel actions, including actions directed at a live human participant. Longer existing logs provide descriptive evidence of repeated ties and distinct relational trajectories; the preregistered intervention identifies the next evidentiary step. The central evaluation question is: “What did this world enable its inhabitants to do together?”

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